Place Value and Powers of 10

Name _____ **Date** _____ No calculator Version 1

KEY WORDS digit · place value · million · power of 10 · exponent · expanded form

A Place value in large numbers ●○○ — ANSWER ON THE SHEET

Remember: The place of a digit tells you its value. In 3,482,917 the 4 is in the hundred thousands place, so it is worth 400,000.

1. Write the value of the underlined digit.

a) $\underline{4},582$ _____ **b)** $3\underline{6},907$ _____ **c)** $\underline{2}18,450$ _____

d) $\underline{5},760,312$ _____ **e)** $90\underline{4},615$ _____ **f)** $1,\underline{2}83,590$ _____

g) $72,\underline{8}46$ _____ **h)** $6,0\underline{4}5,200$ _____ **i)** $3,91\underline{7},484$ _____

B Powers of 10 ●○○ — ANSWER ON THE SHEET

Remember: The power tells you how many zeros follow the 1: $10^3 = 10 \times 10 \times 10 = 1{,}000$.

2. Write each power of 10 as an ordinary number.

a) 10^2 = _____ **b)** 10^3 = _____ **c)** 10^1 = _____

d) 10^5 = _____ **e)** 10^4 = _____ **f)** 10^6 = _____

3. Write each number as a power of 10.

a) 1,000 = _____ **b)** 100 = _____ **c)** 100,000 = _____

d) 10 = _____ **e)** 1,000,000 = _____ **f)** 10,000 = _____

C Expanded form ●●○ — WORK IN YOUR BOOK

Remember: Each digit is multiplied by the power of 10 for its place: $4{,}205 = 4 \times 10^3 + 2 \times 10^2 + 5 \times 10^0$. A place that holds a zero is left out.

4. Write as an ordinary number.

a) $4 \times 10^1 + 7 \times 10^0$

b) $3 \times 10^2 + 6 \times 10^1 + 2 \times 10^0$

c) $5 \times 10^3 + 8 \times 10^1$

d) $2 \times 10^4 + 4 \times 10^2 + 9 \times 10^0$

e) $3 \times 10^5 + 7 \times 10^2$

f) $6 \times 10^6 + 1 \times 10^3 + 5 \times 10^1$

5. Write each number in expanded form, using powers of 10.

a) 58

b) 731

c) 4,206

d) 90,350

e) 108,000

f) 3,050,400

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

D Comparing and ordering ●●○ — ANSWER ON THE SHEET

> **Remember:** Compare the digits from the left. A number with more digits is always larger.

6. Write <, > or = in each box.

a) $4{,}520 \square 4{,}502$ **b)** $98{,}765 \square 100{,}234$ **c)** $360{,}000 \square 306{,}000$

d) $10^3 \square 1{,}000$ **e)** $9{,}999 \square 10^4$ **f)** $10^5 \square 50{,}000$

g) $1{,}200{,}000 \square 10^6$ **h)** $10^2 \square 10^3$

7. Write each list in order, from smallest to largest.

a) 4,310, 4,130, 4,031, 4,301 _____

b) 65,200, 62,500, 65,020, 56,200 _____

c) 1,040,000, 1,004,000, 1,400,000, 140,000 _____

E Multiplying and dividing by 10, 100 and 1,000 ●●○ — ANSWER ON THE SHEET

> ⚠ **Watch out:** Do not just "add a zero". Every digit moves: one place for 10, two places for 100 and three places for 1,000. So $3.7 \times 10 = 37$, not 3.70.

8. Work out.

a) 45×100 = _____ **b)** 3.7×10 = _____ **c)** $0.62 \times 1{,}000$ = _____ **d)** 208×10 = _____

e) $4{,}800 \div 100$ = _____ **f)** $56 \div 10$ = _____ **g)** $7 \div 100$ = _____ **h)** $930 \div 1{,}000$ = _____

i) 1.05×100 = _____ **j)** 0.4×10 = _____ **k)** $2.5 \div 10$ = _____ **l)** $69 \div 100$ = _____

★ Extension — FOR EARLY FINISHERS

9. How many times larger is the value of the 7 in 7,300 than the value of the 7 in 73?

10. Find n if $10^3 \times 10^2 = 10^n$.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

Place Value and Powers of 10

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS digit · place value · million · power of 10 · exponent · expanded form

A Place value in large numbers ●○○ — ANSWER ON THE SHEET

Remember: The place of a digit tells you its value. In 3,482,917 the 4 is in the hundred thousands place, so it is worth 400,000.

1. Write the value of the underlined digit.

a) $\underline{7},265$ _____ **b)** $5\underline{8},304$ _____ **c)** $\underline{4}31,720$ _____

d) $\underline{8},250,961$ _____ **e)** $60\underline{3},847$ _____ **f)** $2,\underline{4}91,350$ _____

g) $96,\underline{5}18$ _____ **h)** $7,0\underline{3}2,600$ _____ **i)** $5,82\underline{6},193$ _____

B Powers of 10 ●○○ — ANSWER ON THE SHEET

Remember: The power tells you how many zeros follow the 1: $10^3 = 10 \times 10 \times 10 = 1{,}000$.

2. Write each power of 10 as an ordinary number.

a) 10^3 = _____ **b)** 10^2 = _____ **c)** 10^4 = _____

d) 10^1 = _____ **e)** 10^6 = _____ **f)** 10^5 = _____

3. Write each number as a power of 10.

a) 100 = _____ **b)** 10,000 = _____ **c)** 1,000 = _____

d) 10 = _____ **e)** 100,000 = _____ **f)** 1,000,000 = _____

C Expanded form ●●○ — WORK IN YOUR BOOK

Remember: Each digit is multiplied by the power of 10 for its place: $4{,}205 = 4 \times 10^3 + 2 \times 10^2 + 5 \times 10^0$. A place that holds a zero is left out.

4. Write as an ordinary number.

a) $8 \times 10^1 + 3 \times 10^0$

b) $5 \times 10^2 + 9 \times 10^1 + 4 \times 10^0$

c) $7 \times 10^3 + 2 \times 10^1$

d) $4 \times 10^4 + 6 \times 10^2 + 7 \times 10^0$

e) $5 \times 10^5 + 3 \times 10^2$

f) $2 \times 10^6 + 4 \times 10^3 + 7 \times 10^1$

5. Write each number in expanded form, using powers of 10.

a) 74

b) 926

c) 5,308

d) 60,720

e) 205,000

f) 4,070,100

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

D Comparing and ordering ●●○ — ANSWER ON THE SHEET

> **Remember:** Compare the digits from the left. A number with more digits is always larger.

6. Write <, > or = in each box.

a) $6{,}730 \square 6{,}703$ **b)** $87{,}654 \square 100{,}123$ **c)** $540{,}000 \square 504{,}000$

d) $10^2 \square 100$ **e)** $999 \square 10^3$ **f)** $10^4 \square 40{,}000$

g) $1{,}500{,}000 \square 10^6$ **h)** $10^5 \square 10^4$

7. Write each list in order, from smallest to largest.

a) 7,520, 7,250, 7,052, 7,502 _____

b) 83,400, 84,300, 83,040, 38,400 _____

c) 2,060,000, 2,006,000, 2,600,000, 260,000 _____

E Multiplying and dividing by 10, 100 and 1,000 ●●○ — ANSWER ON THE SHEET

> ⚠ **Watch out:** Do not just "add a zero". Every digit moves: one place for 10, two places for 100 and three places for 1,000. So $3.7 \times 10 = 37$, not 3.70.

8. Work out.

a) 63×100 = _____ **b)** 5.2×10 = _____ **c)** $0.48 \times 1{,}000$ = _____ **d)** 307×10 = _____

e) $6{,}500 \div 100$ = _____ **f)** $84 \div 10$ = _____ **g)** $9 \div 100$ = _____ **h)** $710 \div 1{,}000$ = _____

i) 2.06×100 = _____ **j)** 0.7×10 = _____ **k)** $3.5 \div 10$ = _____ **l)** $48 \div 100$ = _____

★ Extension — FOR EARLY FINISHERS

9. How many times larger is the value of the 5 in 5,200 than the value of the 5 in 52?

10. Find n if $10^4 \times 10^2 = 10^n$.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.